

Hybrid Image Colorization via Regression and Classification

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1. Introduction

Image colorization is the task of predicting the color values of each pixel in the grayscale input image. In this project, images are represented in the CIELAB colorspace, where the L channel represents the luminance (the grayscale), and the (a, b) channels represent the chrominance (the color).

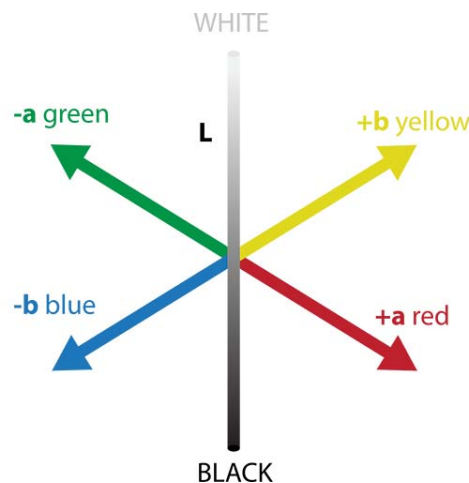


Figure 1: CIELAB colorspace.

Using this decomposition, the model takes input L and predicts (a, b) . Intuitively, the color prediction can be done using regression and classification approaches. In particular:

- **Regression:** Predict continuous (a, b) values. This approach is simple, but it can produce desaturated colors because of averaging effects.
- **Classification:** Discretize the (a, b) space into bins and predict a probability distribution over them. This method can generate vivid colors, but incorrect bin predictions can lead to unrealistic results.

This project explores and compares both methods using the U-Net architecture. In addition, motivated by limitations of each individual approach, this project proposes a hybrid approach that uses the color bins learned from classification and improves them with a more precise regression output.

2. Objectives

The objectives of this project are:

- Implement and compare regression- and classification-based image colorization using the U-Net architecture.
- Develop a hybrid approach using the U-Net architecture and evaluate its performance relative to individual approaches.

3. State-of-the-Art Methods in Image Colorization

One of the prominent works in image colorization is *Let There Be Color!* by Iizuka et al. [1]. They proposed a convolutional neural network with a fusion layer that extracts global and local features and fuses them. The model takes the luminance L channel as input and predicts continuous (a, b) chrominance values. Afterward, L and (a, b) channels are combined to produce the final colorized image.

When Generative Adversarial Networks (GANs) were introduced, adversarial training became a powerful image colorization approach. A representative example is *ChromaGAN* [2]. There, the generator predicts pixel color using semantic cues, while the discriminator ensures that the predicted colors follow class-consistent distributions.

More recently, diffusion-based methods have established a new performance benchmark for image-to-image tasks. One such model is *Palette*, which uses a U-Net and a denoising diffusion process to transform an input image into the desired output [3]. In particular, the model takes the grayscale input image and attaches it to the noisy intermediate images during diffusion. After diffusion, the model gradually denoises them, producing high-quality and realistic outputs.

Transformer architectures have advanced the field along with diffusion methods. The first fully self-attention-based colorization model, *ColTran* [4], was a significant step forward in image colorization. Specifically, it generates a low-resolution colored image with an autoregressive transformer and improves it using transformer upsamplers. Since every layer uses the grayscale input as context, the model learns the structure of the scene and performs better colorization than earlier methods.

4. Methodology

4.1 Dataset and Preprocessing

In this project, the Tiny ImageNet dataset, which consists of RGB images resized to 64×64 pixels, was used. A total of 24,000 images were used for training and 6,000 images were used for testing. Each image was converted from the RGB colorspace to the CIELAB colorspace.

During preprocessing, the L channel was normalized to the range $[0, 1]$, while the (a, b) channels were normalized to $[-1, 1]$. All models received the L channel as input and was trained to predict the corresponding (a, b) channels for colorization. In addition, classification U-Net and hybrid U-Net required discretizing the (a, b) channels into 64 bins before training.

4.2 Model Architecture and Training

The two U-Net-based architectures used for image colorization are a regression model that predicts continuous (a, b) values, and a classification model that predicts discretized color bins. Both models work with grayscale input images of size $64 \times 64 \times 1$ and share the same U-Net backbone. The only difference is the design of the final output layer.

The shared architecture is as follows:

- **Encoder**

- Level 1: $\text{Conv}(1 \rightarrow 64) \rightarrow \text{Conv}(64 \rightarrow 64), \text{MaxPool}(2)$

- Level 2: Conv(64→128) → Conv(128→128), MaxPool(2)
- Level 3: Conv(128→256) → Conv(256→256), MaxPool(2)

- **Bottleneck**

- Conv(256→512) → Conv(512→512)

- **Decoder**

- Level 3: UpSample(2) + skip(c3), Conv(512→256)
- Level 2: UpSample(2) + skip(c2), Conv(256→128)
- Level 1: UpSample(2) + skip(c1), Conv(128→64)

The distinction between the two models is in the output layer:

- **Regression Output:** A 1×1 convolution with two filters and a `tanh` activation,

$$\text{Conv2D}(2, 1 \times 1, \text{activation} = \tanh).$$

It produces continuous normalized (a, b) values in $[-1, 1]$.

- **Classification Output:** A 1×1 convolution with N_{bins}^2 filters and a softmax activation,

$$\text{Conv2D}(N_{\text{bins}}^2, 1 \times 1, \text{activation} = \text{softmax}).$$

It produces a per-pixel probability distribution over discretized color bins.

Different colors can have the same luminance, causing regression models to average possible color values and create a “sepia” effect. The hybrid approach solves this by combining classification for color family with regression for refinement. The hybrid approach uses the same U-Net backbone as regression and classification. However, as above, the difference is in the final layer. It consists of two heads: a **classification head** and a **regression head**. The final chrominance is computed as

$$(\hat{a}, \hat{b}) = (a, b)_{\text{class}} + \Delta(a, b)_{\text{reg}},$$

where

$$(a, b)_{\text{class}} = \text{dequantize}\left(\arg \max_k p_{\theta}(k \mid L)\right),$$

and $\Delta(a, b)_{\text{reg}}$ is the regression head output.

Training was done on a Kaggle Notebook using an NVIDIA T4 GPU to accelerate the process. Both models were trained end-to-end using the Adam optimizer (default learning rate $1e-3$) for 70 epochs with a mini-batch size of 32. However, the regression U-Net was trained using mean squared error, whereas the classification U-Net used sparse categorical cross-entropy.

Thus, by using an identical U-Net backbone and a similar training setup, the regression and classification models can be fairly compared.

The hybrid model was trained for 50 epochs using a joint loss:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{reg}},$$

where $\mathcal{L}_{\text{cls}}$ is the sparse categorical cross-entropy loss, $\mathcal{L}_{\text{reg}}$ is the mean squared error loss, and $\lambda = 0.3$ is the contribution of the regression loss. The classification loss is the main objective because it determines the color family. In contrast, the regression loss is weighted to ensure that it does not overpower the primary classification decision.

5. Results

Figures 2, 3, and 4 provide the training loss curves for the regression, classification, and hybrid models, respectively.

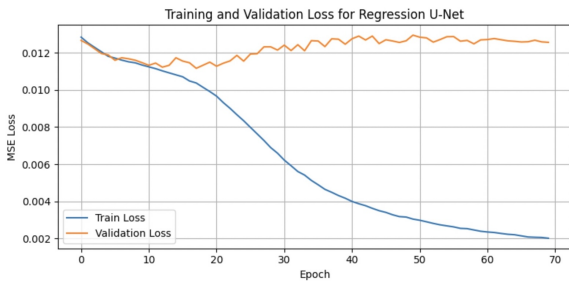


Figure 2: Regression loss.

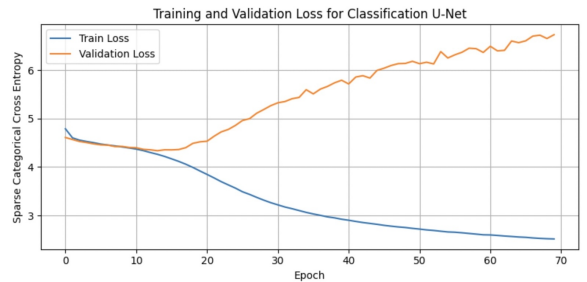


Figure 3: Classification loss.

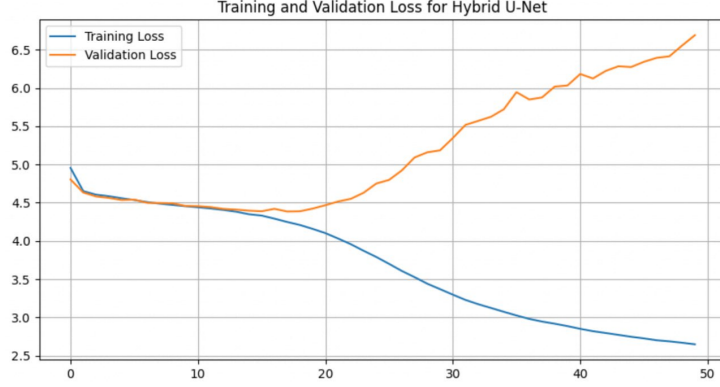


Figure 4: Hybrid loss.

	1	2	3	4	5	6	7	8	9	10
MSE_reg	0.0014	0.0027	0.0006	0.0029	0.0322	0.0059	0.0071	0.0219	0.0015	0.0017
SSIM_reg	0.9832	0.9298	0.9893	0.9596	0.8148	0.9433	0.9449	0.8796	0.9687	0.9493
MSE_class	0.0024	0.0052	0.0014	0.0050	0.0344	0.0076	0.0053	0.0244	0.0020	0.0025
SSIM_class	0.9558	0.8172	0.9810	0.9326	0.7683	0.9241	0.9400	0.8230	0.9574	0.9316
MSE_hybrid	0.0026	0.0051	0.0007	0.0044	0.0258	0.0053	0.0165	0.0237	0.0034	0.0018
SSIM_hybrid	0.9685	0.8800	0.9896	0.9495	0.8369	0.9511	0.8870	0.8514	0.9574	0.9479

Table 1: MSE and SSIM results for Regression, Classification, and Hybrid Methods.

To evaluate the performance of different approaches, 10 random test images were used for colorization. Although the main assessment was based on qualitative analysis, quantitative metrics were also compared. In particular, the mean squared error (MSE) and the structural similarity index (SSIM) were computed. The SSIM between two images x and y is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (1)$$

where μ_x, μ_y are mean intensities, σ_x^2, σ_y^2 are variances (contrast), σ_{xy} is covariance (structural similarity), and C_1, C_2 are small constants to stabilize the expression. The SSIM can range from -1 to 1 , with 1 indicating perfect similarity.

6. Discussion

Even though it was initially assumed that the regression U-Net would produce desaturated results, it actually performed better than expected. Moreover, it trained in a little over one hour, which is considerably faster than the classification U-Net and hybrid U-Net that took three to four training hours. Besides that, both qualitatively and quantitatively,

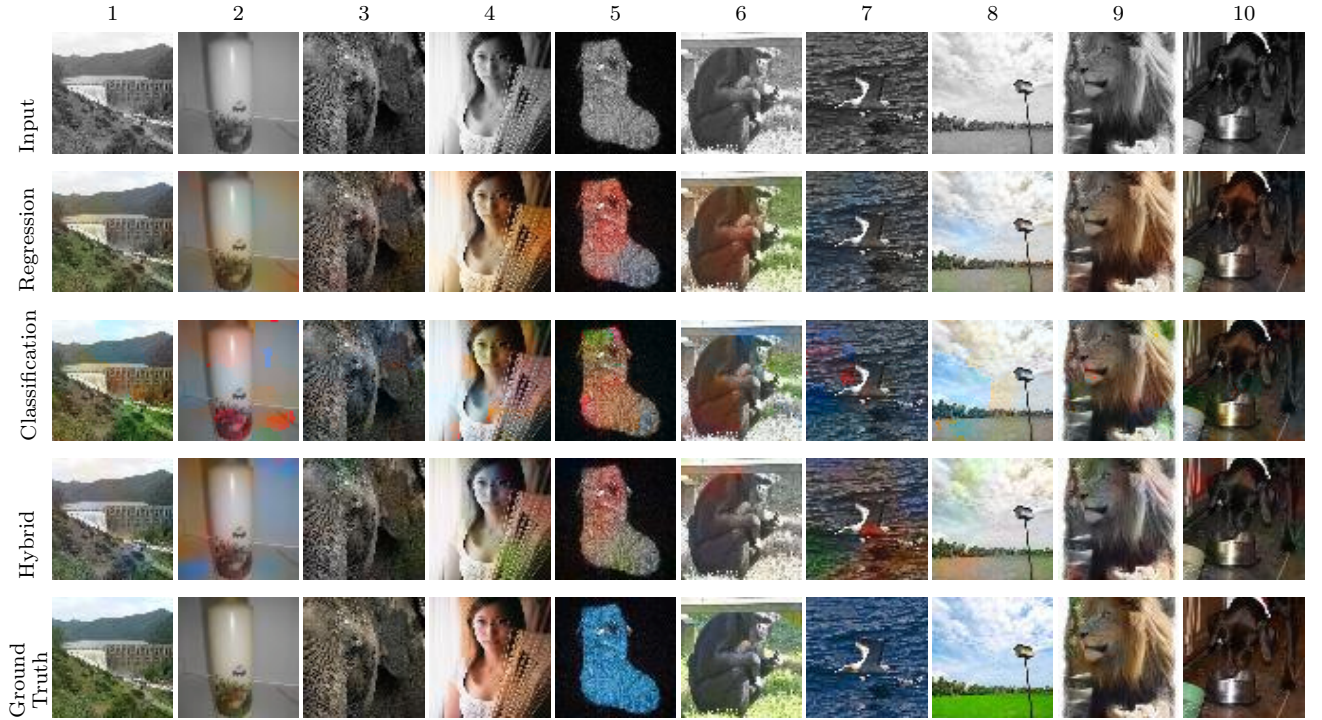


Figure 5: Results.

regression outperformed classification and hybrid approaches in terms of color consistency. Visually, classification tends to predict random colors for some pixels, which leads to inconsistent colorization. Nevertheless, a closer look reveals that classification chooses bolder and more saturated colors. The hybrid approach attempted bolder colors as well, yet the final results look less harsh than those of classification, which supports the proposed idea.

In order to improve overall performance, it is recommended to increase the number of training images and the number of epochs. Although the training plots suggest that additional epochs may not reduce the MSE (since it stops improving at a certain point), the ablation study showed that more epochs introduce more colors. Besides this, the question of whether MSE or SSIM should even be considered representative metrics for colorization requires separate investigation. Without contextual clues from the user, the goal of colorization is to produce realistic colors, not necessarily accurate ones.

Regarding the classification and hybrid U-Net, using a higher number of color bins is expected to improve color richness. However, increasing the number of bins leads to increased training time, so there is a trade-off between accuracy and time. It is also

advised to attempting different the regression and classification loss weights in hybrid loss.

In this project, the U-Net architecture was shallow to provide faster training. Increasing the number of downsampling and upsampling layers would likely improve the model’s overall performance.

In conclusion, due to the fast training and realistic results, the regression U-Net is the obvious winner in this experiment.

Supplementary Materials

The full code implementation is available at:

<https://github.com/nrmtkz/image-colorization>.

References

- [1] S. Iizuka, E. Simo-Serra, and H. Ishikawa, “Let there be color!: Joint end-to-end learning of global and local image priors for automatic image colorization with simultaneous classification,” *ACM Transactions on Graphics*, vol. 35, no. 4, pp. 1–11, 2016.
- [2] P. Vitoria, L. Raad, and C. Ballester, “ChromaGAN: Adversarial picture colorization with semantic class distribution,” *arXiv preprint arXiv:1907.09837*, 2019.
- [3] C. Saharia *et al.*, “Palette: Image-to-image diffusion models,” in *SIGGRAPH 2022 Conference Proceedings*, 2022.
- [4] M. Kumar, D. Weissenborn, and N. Kalchbrenner, “Colorization transformer,” in *International Conference on Learning Representations (ICLR)*, 2021.